NEECA
National Energy Efficiency & Conservation Authority



Pakistan is undergoing a significant shift towards urbanization, with millions of people expected to live in urban centers by 2023. This rapid urban growth means a surge in construction activities, offering a unique opportunity to embrace energy-efficient appliances and effective Energy Efficient Building Codes. The traditional approach of horizontal urbanization, spreading across vast areas of land, also raises concerns about the efficient use of resources, particularly agricultural land. By prioritizing energy efficiency, we can minimize energy consumption, reduce greenhouse gas emissions, and create a healthier living for all.

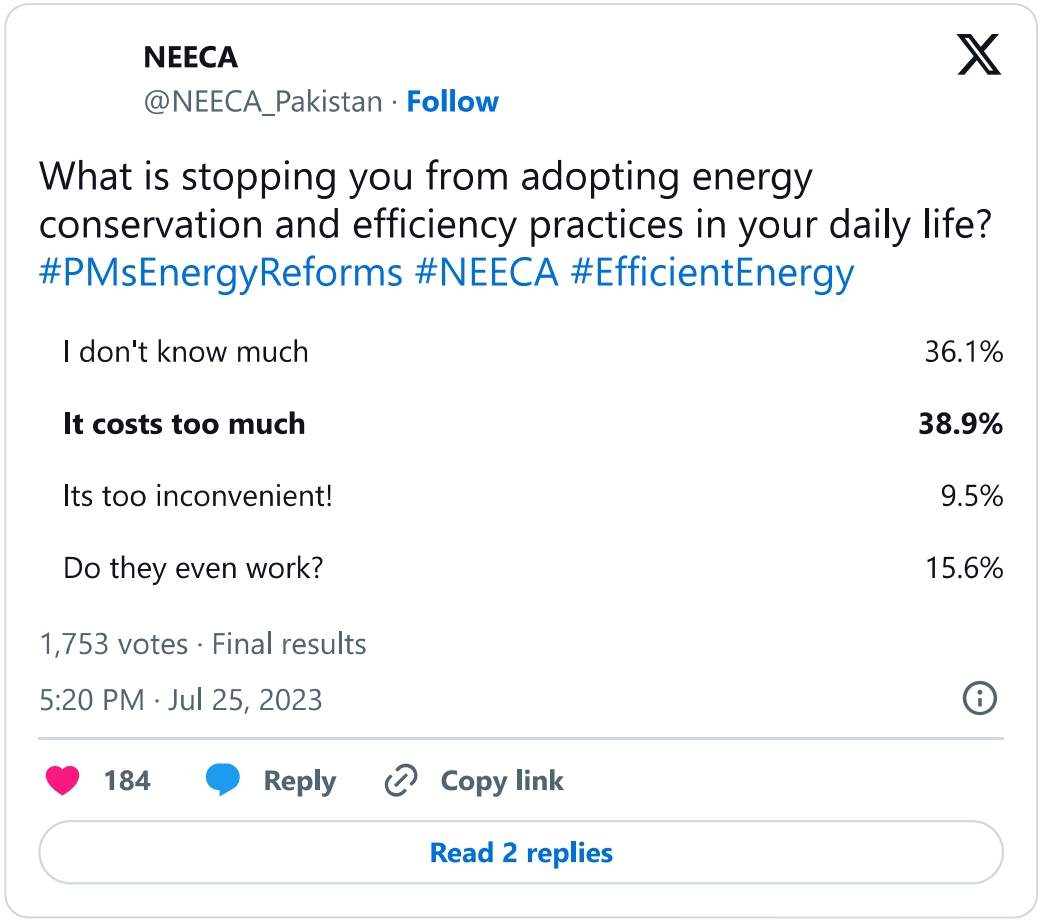
The National Energy Efficiency and Conservation Authority (NEECA) is committed to transforming the building industry by promoting energy efficiency practices. Let's explore how NEECA is leading the way in creating comfortable, cost-effective, and eco-friendly homes and why it matters for you.

NEECA's Solutions: Energy-Efficient Homes for All

NEECA recognizes the importance of energy efficiency in the building sector and has introduced a range of initiatives to promote energy-smart practices. One key initiative is the implementation of the Energy Conservation Building Codes (ECBC-2023) and its integration into the building bylaws of provincial development authorities, ensuring energy-efficient practices in new construction as well as in the old buildings through retrofitting techniques mentioned in ECBC-2023. These standards cover various aspects, including building envelopes, heating, ventilation, air conditioning (HVAC) equipment, and lighting. By adhering to these codes, we can ensure that new constructions are energy efficient from the ground up.

To ensure a sustainable and efficient housing solution, it is imperative that we redirect our attention from horizontal urbanization to embracing vertical urbanization. This approach not only helps us preserve agricultural land but also improves energy efficiency and conservation potential in the building sector. By optimizing land usage, preserving agricultural land, and incorporating smarter and energy-efficient housing practices, we can build resilient, livable cities that meet the needs of present and future generations.

NEECA is also working on retrofitting existing buildings with energy-efficient technologies and the detailed design of the retrofitting techniques have been included in the Energy Conservation Building Code 2023. By adopting energy conservation practices and upgrading appliances, we can significantly reduce energy consumption in residential and commercial buildings. This is crucial as buildings currently account for a large portion of energy consumption in Pakistan.



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Reference code:

Q&A with Khalil Raza and Zeeshan Ullah from NEECA

Your Role in Building a Better Future:

You have the power to make a difference in building a better future. Start by adopting energy-efficient practices in your home. Insulate your homes and buildings, maximize the use of natural ventilation and lighting, upgrade to energy-efficient appliances, implement smart energy management systems, and consider retrofitting your home with energy-saving technologies. You can also take the first step by supporting NEECA's initiatives and advocate for energy efficiency in your community.

Together, let's build an energy-smart future where energy-efficient homes are the norm. By taking small steps towards energy efficiency, we can create an energy smart and more prosperous Pakistan for generations to come.

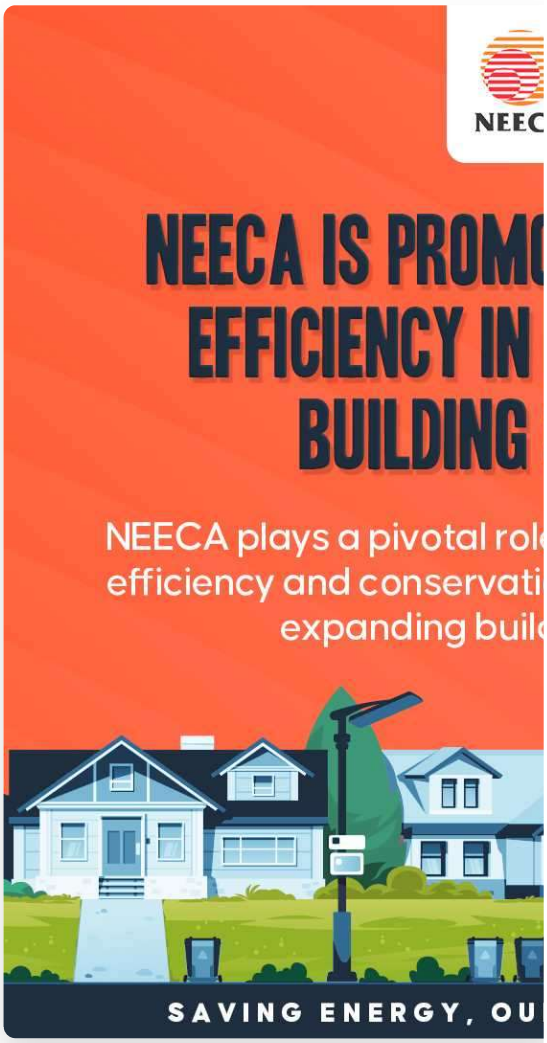
> "Pakistan succeeded in creating a national roadmap for Energy Conservation, which directly impacted the Country’s Foreign exchange."
>
> — Mr. Agha Hassan Baloch


NEECA
MAKE YOUR HOME ENERGY-EFFICIENT!
Upgrade insulation, seal duct leaks, and use energy-saving power strips for maximum impact!


NEECA
UNLOCK THE POWER OF EFFICIENT BUILDING DESIGN
By integrating proper orientation, insulation, and efficient materials and windows, an energy-efficient building can reduce energy consumption by up to 30% compared to conventional designs!

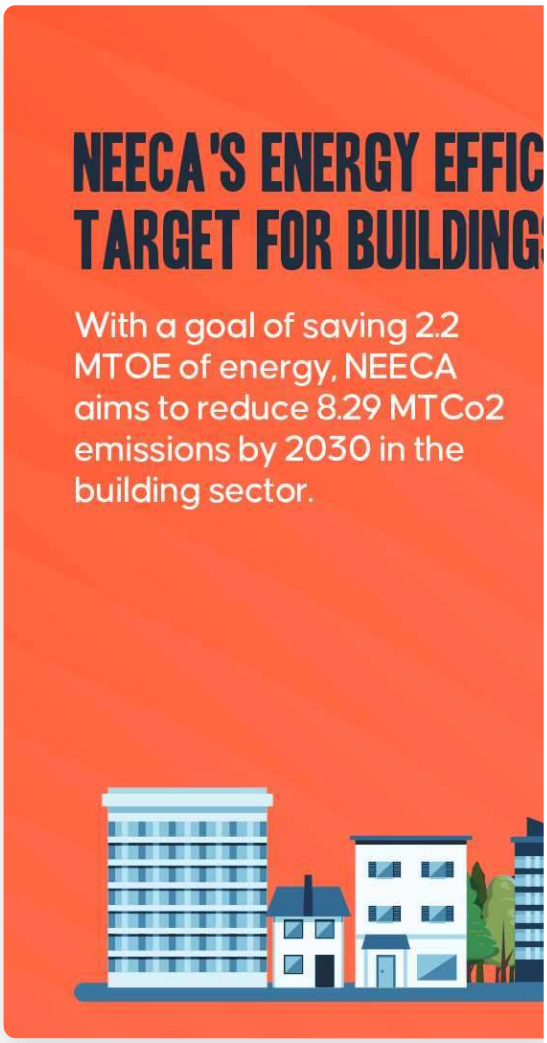

NEECA
HERE IS WHAT NEECA HAS PLANNED TO SAVE ENERGY IN THE BUILDING INDUSTRY:
Introducing the building rating system on the basis of energy efficiency and conservation design and construction techniques mentioned in the Energy Conservation Building Code 2023
Introduced Minimum Energy Performance Standards (MEPS) and Mandatory Energy Labels for electric fans, ACs, refrigerators, lights and electric motors
On-bill financing mechanisms for the provision of energy-efficient home appliances
Source: NEECA Strategic Plan 2020-2025


NEECA
SWITCH TO EFFICIENT APPLIANCES, SAVE ENERGY
Switching from incandescent to LED bulbs can save up to 80% in energy costs!



NEECA IS PROMOTING E

NEECA plays a pivotal role in advoc
conservation in Pakistan's rapidly e
retrofitting of existing buildings with
and adoption of energy conservatio
energy savings given that residentia
22.2 % of final energy consumption
emissions.



NEECA'S ENERGY EFFICI

With a goal of saving 2.2 MTOE of e
8.29 MTC02 emissions by 2030 in t
buildings must adhere to Energy Co
2024, while older buildings will unde
with five-year energy-saving plans l
performance standards for applianc
2023, and public procurement guide
labeled products.

The building Sector of Pakistan has massive potential for Energy savings.

THE POWER OF ENERGY EFFICIENT BUILDINGS:

Energy Efficiency Tips for Residential Stakeholders



NEECA

The building sector in Pakistan has a massive potential for energy savings, estimated to be around 30% through the implementation of energy-efficient measures and technologies.

The energy savings achieved can translate into reduced energy bills, improved indoor air quality, thermal comfort, enhanced financial sustainability and improved overall economic competitiveness.

Prominent Challenges

- Lack of widespread adoption and enforcement of Energy Conservation Building Codes and standards during the design, construction and operation phases.
- High upfront costs for retrofits in existing buildings act as a barrier for implementing energy-efficient practices in buildings.
- The current practice of constructing a new building often lacks expertise in economical design and energy efficiency, resulting in wastage of resources.
- Absence of regular energy audits and performance assessments, making it difficult to measure the impact of energy efficiency measures and identify areas for improvement.
- Shortage of skilled professionals in energy-efficient building design and practices, such as architects, energy engineers, and contractors.

Potential Solutions

- Conduct Energy Audits of your building envelope (insulation and air leakages) and your appliances.
- Implement Energy Conservation Building Code (ECBC-2023) to address energy inefficiencies in existing or new buildings.
- Opt for energy efficient building materials, windows and doors in compliance with ECBC-2023.
- Invest in rainwater harvesting and water reuse design for your building.
- Retrofit buildings with improved passive design, insulation and shading strategies as recommended by ECBC 2023.
- Design or retrofit buildings with improved envelope and ensure insulation in compliance with ECBC-2023.



Discover the rewards of energy efficiency, choose efficient appliances and use energy wisely, upgrade for efficiency, and inspire others to join the energy efficiency movement.

NEECA is committed to saving 2.2 MTOE of energy in the building sector which will result in emission reduction of 8.29 MTCo2 by 2030 and we need your help to do it!

Source: NEECA Strategic Plan, Research and Reports

Energy Savings translate into reduced energy bills.

NEECA
National Energy Efficiency & Conservation Authority

Leading Pakistan's energy efficiency and conservation initiatives for a sustainable and energy-secure future.

Quick Links

- Special Investment Facilitation Council
- Ministry of Energy (Power Division)
- Energy Department-Government of the Punjab
- Punjab Energy Efficieny and Conservation Agency (PEECA)
- Sindh Energy Efficiency and Conservation Authority (SEECA)
- Pakhtunkhwa Energy Development Organization(PEDO)
- Energy & Power Department Government of Khyber Pakhtunkhwa
- Energy Department- Government of Balochistan

Sectors We Serve

- Agriculture Sector
- Power & Energy Sector
- Transport Sector
- Industry Sector
- Building & Residential Sector

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